

Intro to ggplot and Data visualization

ggplot, geom_point

YOUR NAME HERE

Learning Goals:

- identify basics of a dataset: rows, columns, variables
- gain familiarity with basic R syntax, tidyverse
- identify a function and its structure (has (), inputs/arguments go inside ())
- make a scatterplot
- describe trends shown in a scatterplot

First, we load the packages we will be using and the data:

```
library(tidyverse)
library(palmerpenguins)
data(penguins)
```

Background

For today's exercise, we will use the **penguins** dataset. Be sure that you have run the chunk above, because "penguins" is from the **palmerpenguins** library.

More details about palmerpenguins [here](#)

We can view the dataset by typing its name:

```
penguins

## # A tibble: 344 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>
## 1 Adelie  Torgersen         39.1          18.7          181          3750
## 2 Adelie  Torgersen         39.5          17.4          186          3800
## 3 Adelie  Torgersen         40.3           18          195          3250
## 4 Adelie  Torgersen          NA           NA           NA           NA
## 5 Adelie  Torgersen         36.7          19.3          193          3450
## 6 Adelie  Torgersen         39.3          20.6          190          3650
## 7 Adelie  Torgersen         38.9          17.8          181          3625
## 8 Adelie  Torgersen         39.2          19.6          195          4675
## 9 Adelie  Torgersen         34.1          18.1          193          3475
## 10 Adelie Torgersen         42           20.2          190          4250
## # i 334 more rows
## # i 2 more variables: sex <fct>, year <int>
```

```
# to see the first 10 rows and all columns
print(penguins, n = 10, width = Inf)
```

```
## # A tibble: 344 x 8
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>
## 1 Adelie  Torgersen         39.1          18.7          181          3750
## 2 Adelie  Torgersen         39.5          17.4          186          3800
## 3 Adelie  Torgersen         40.3           18          195          3250
## 4 Adelie  Torgersen          NA           NA           NA           NA
## 5 Adelie  Torgersen         36.7          19.3          193          3450
## 6 Adelie  Torgersen         39.3          20.6          190          3650
## 7 Adelie  Torgersen         38.9          17.8          181          3625
## 8 Adelie  Torgersen         39.2          19.6          195          4675
## 9 Adelie  Torgersen         34.1          18.1          193          3475
## 10 Adelie Torgersen         42           20.2          190          4250
##   sex      year
##   <fct> <int>
## 1 male   2007
## 2 female 2007
## 3 female 2007
## 4 <NA>   2007
## 5 female 2007
## 6 male   2007
## 7 female 2007
## 8 male   2007
## 9 <NA>   2007
## 10 <NA>  2007
## # i 334 more rows
```

We can view the help file for penguins in this way:

```
?penguins
```

- 1) How many rows are in the dataset? What does each row represent?
- 2) For each of the following variables, indicate whether it is categorical or numerical. If it is categorical, do the categories have a specific order? If it is numerical, is it discrete (whole numbers) or continuous (decimals possible)?

species

island

bill_length_mm

body_mass_g

year

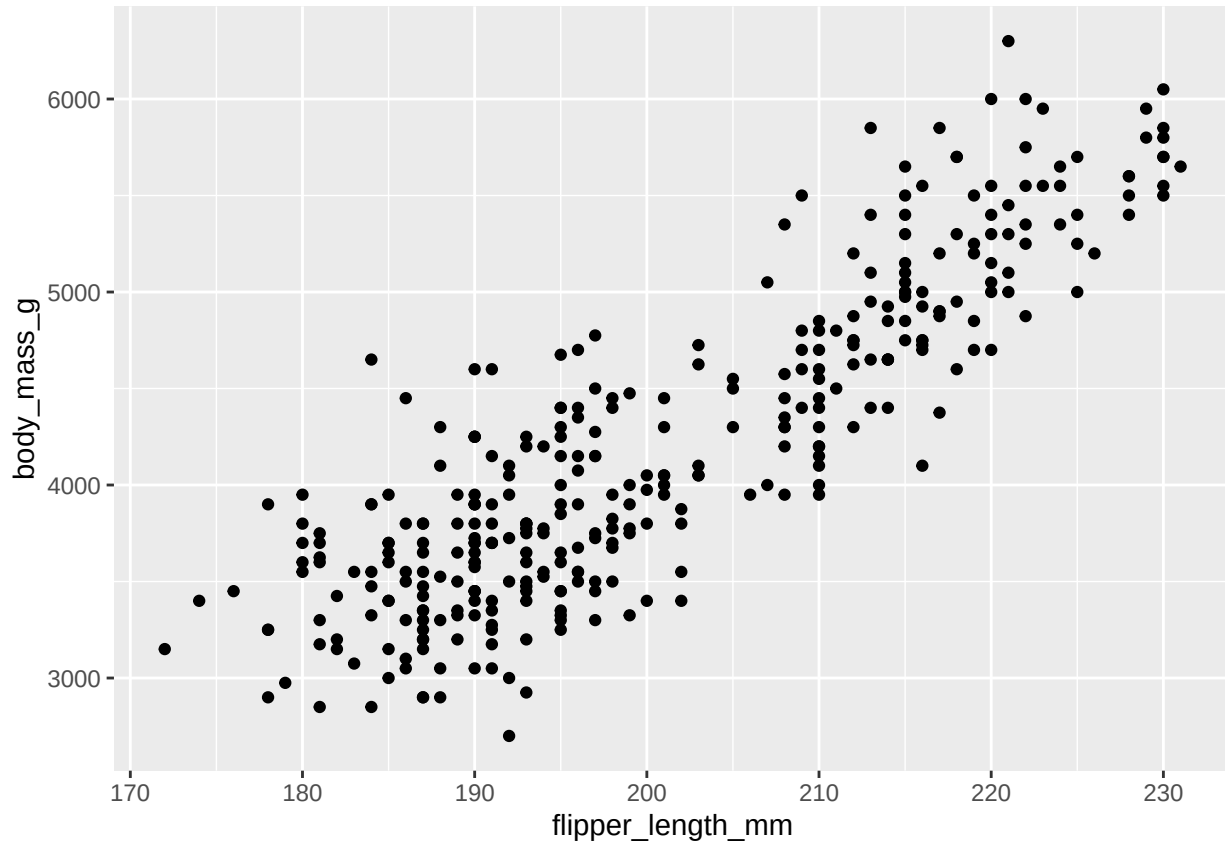
Making a scatterplot with `geom_point()`

To examine the relationship between two numerical variables, we can use a *scatter plot*.

An example plot:

```
ggplot( #LINE 1
  data = penguins, #LINE 2
  mapping = aes(x = flipper_length_mm, y = body_mass_g)) + #LINE 3
  geom_point() #LINE 4
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



3) Identify what each line in the above code does:

Line 1:

Line 2:

Line 3:

Line 4:

4) Read the warning message that appears. What does it mean? Why is important that R notifies you of missing values?

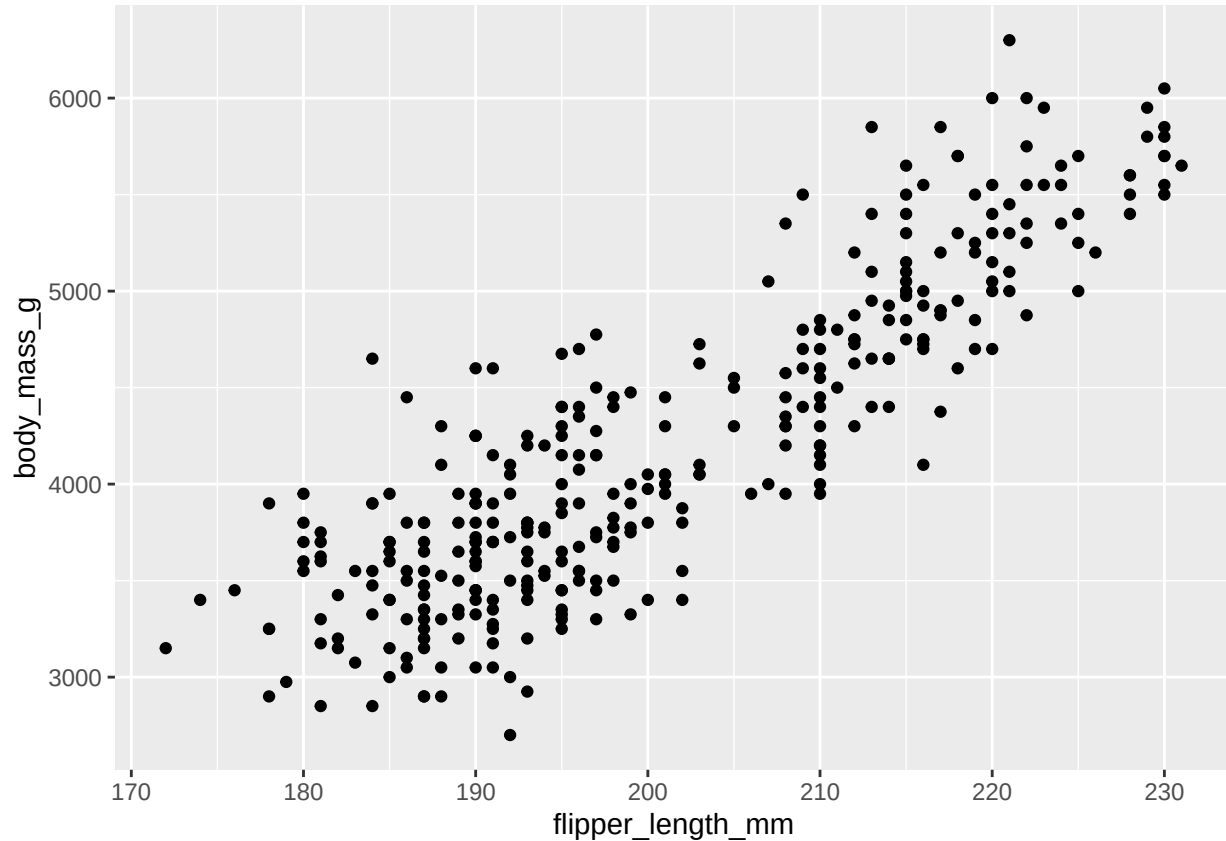
Mapping aesthetics to variables

5) Modify the code below to map the color aesthetic to `species` (i.e. each species of penguin should be a different color).

```
ggplot(
  data = penguins,
```

```
mapping = aes(x = flipper_length_mm, y = body_mass_g)) +  
geom_point()
```

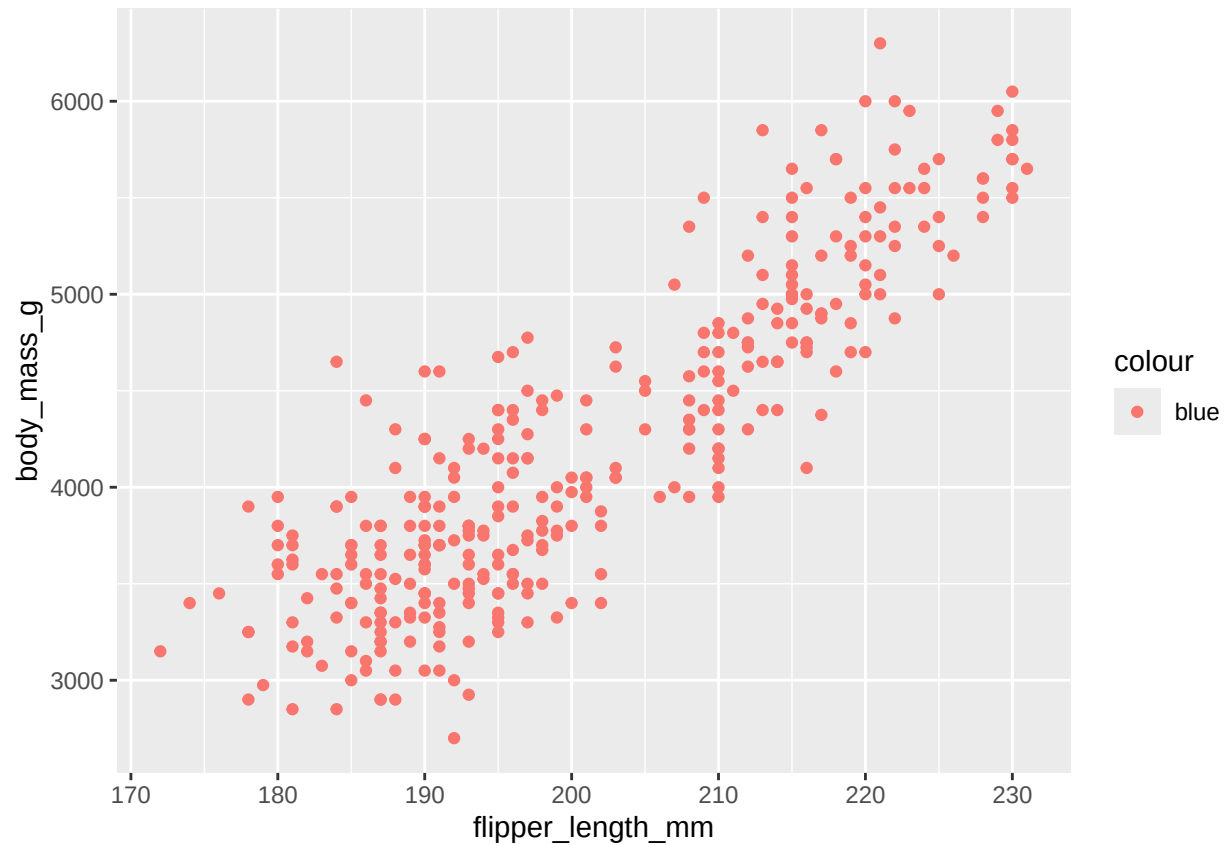
```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



Setting aesthetics to fixed values

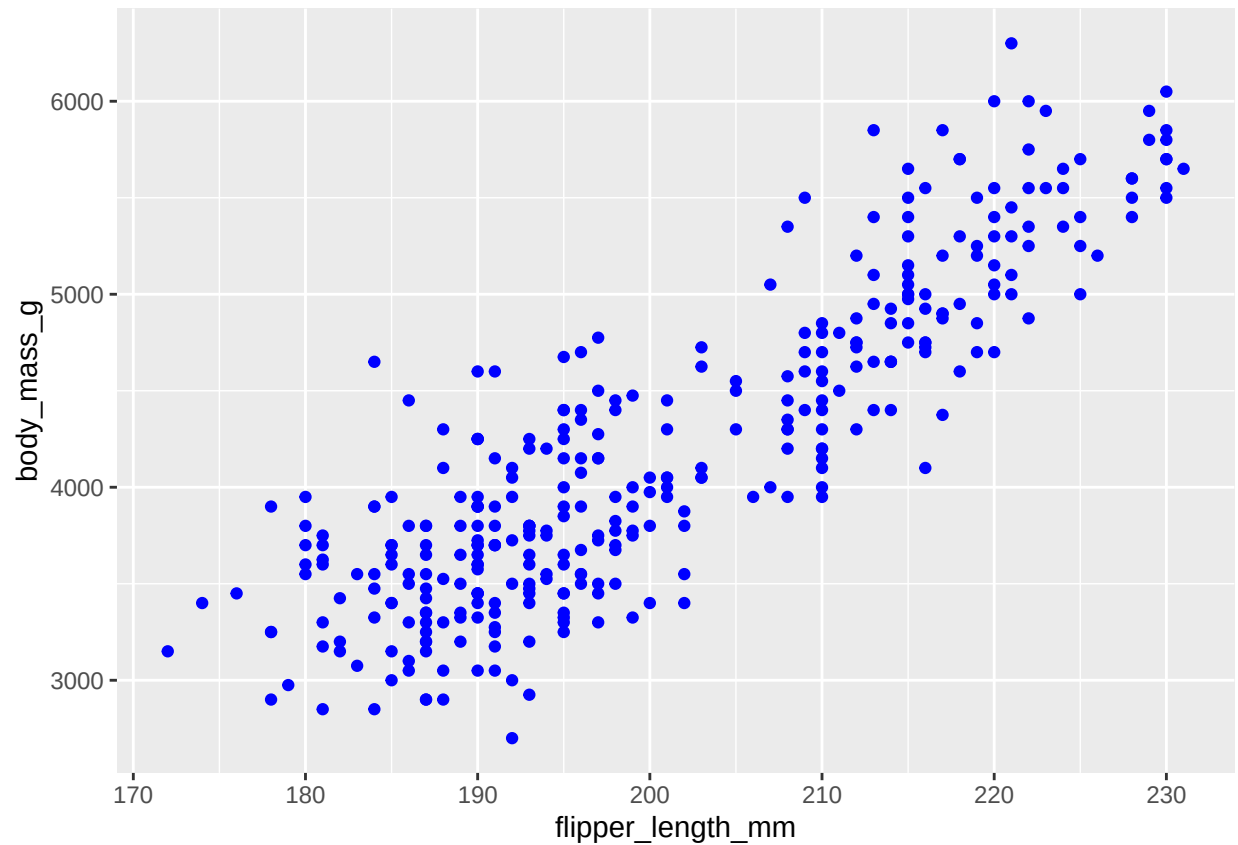
```
ggplot(data = penguins) +  
  geom_point(  
    mapping = aes(  
      x = flipper_length_mm,  
      y = body_mass_g,  
      color = "blue"  
    )  
  )  
)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
ggplot(data = penguins) +  
  geom_point(  
    mapping = aes(  
      x = flipper_length_mm,  
      y = body_mass_g  
    ),  
    color = "blue"  
  )
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

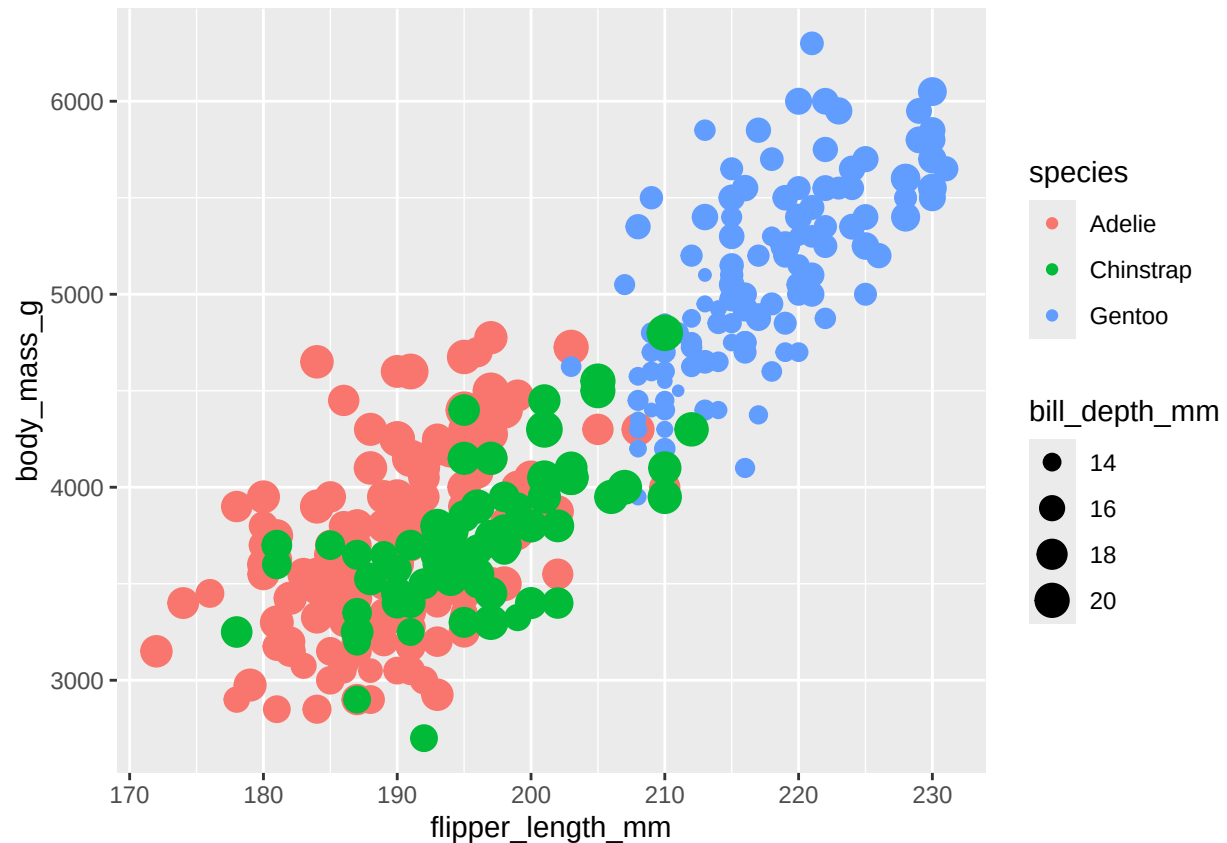


- 6) Explain the difference in the two plots above. Be sure to run the plots one at a time by using “Cmd + Enter”
- 7) When should an aesthetic be inside `aes()` and when should it be outside `aes()`?

Mapping multiple aesthetics

```
ggplot(
  data = penguins,
  mapping = aes(
    x = flipper_length_mm,
    y = body_mass_g,
    color = species,
    size = bill_depth_mm)) +
  geom_point()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



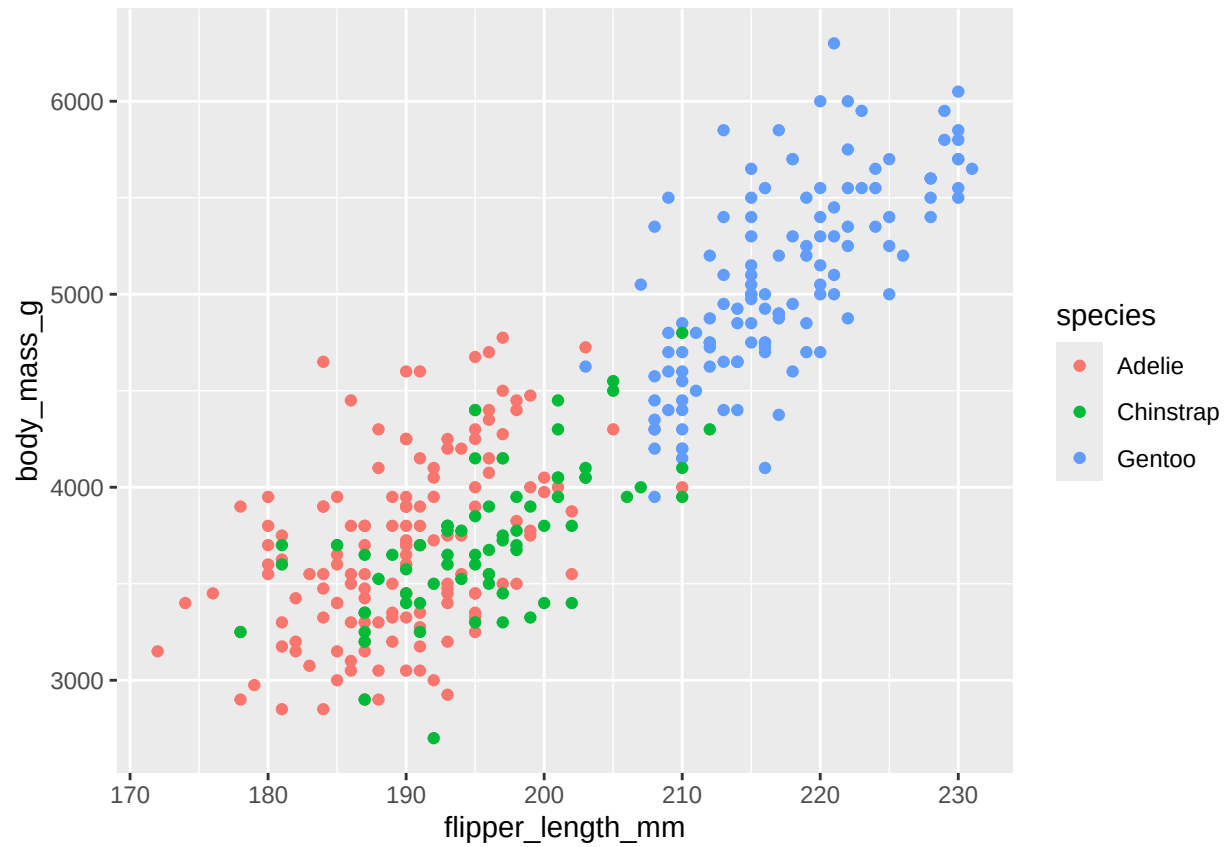
- 8) In the code above, remove `size = bill_depth_mm` and try to instead map shape to species. Add that to the plot above!
- 9) What happens if we *also* map size to species? Do you like the look of this plot?

Shorter code

Note that each of the following produce the exact same graph!!!

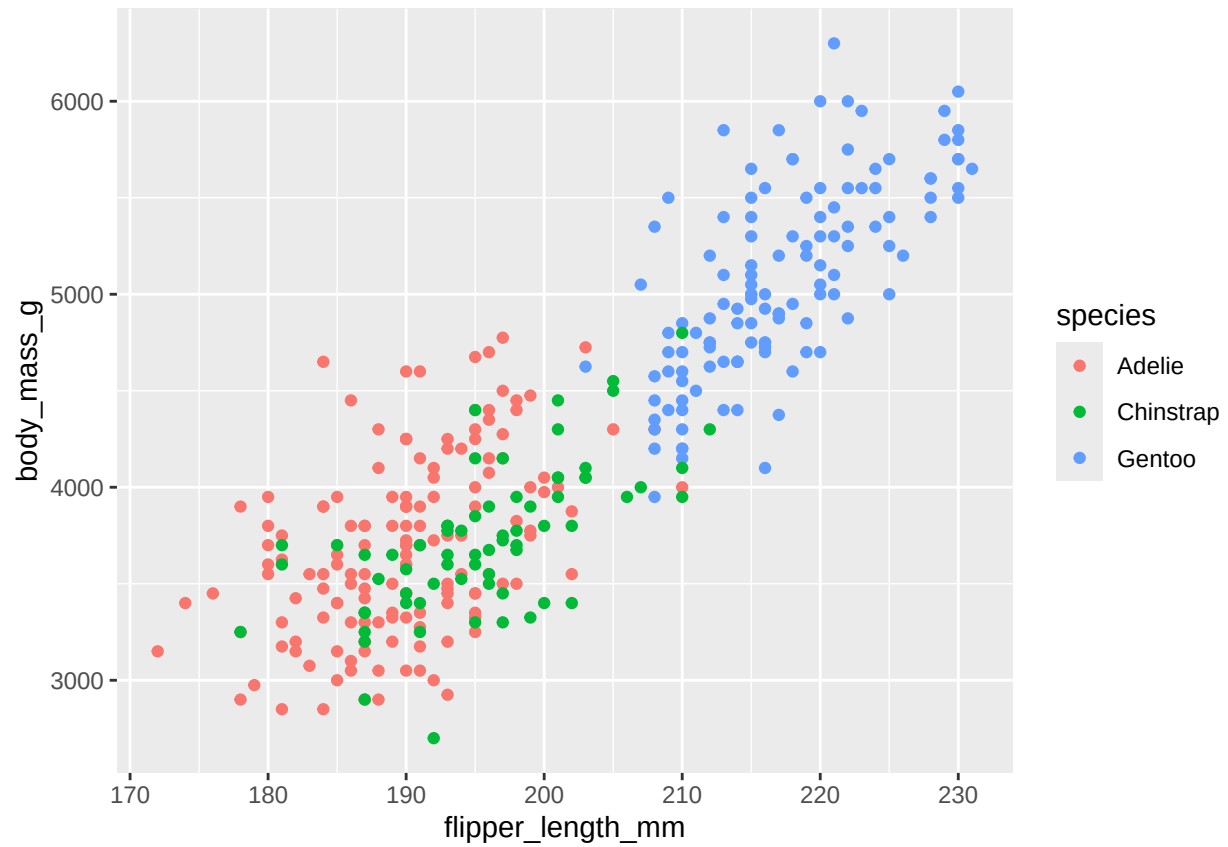
```
ggplot(
  data = penguins,
  mapping = aes(
    x = flipper_length_mm,
    y = body_mass_g,
    color = species)) +
  geom_point()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



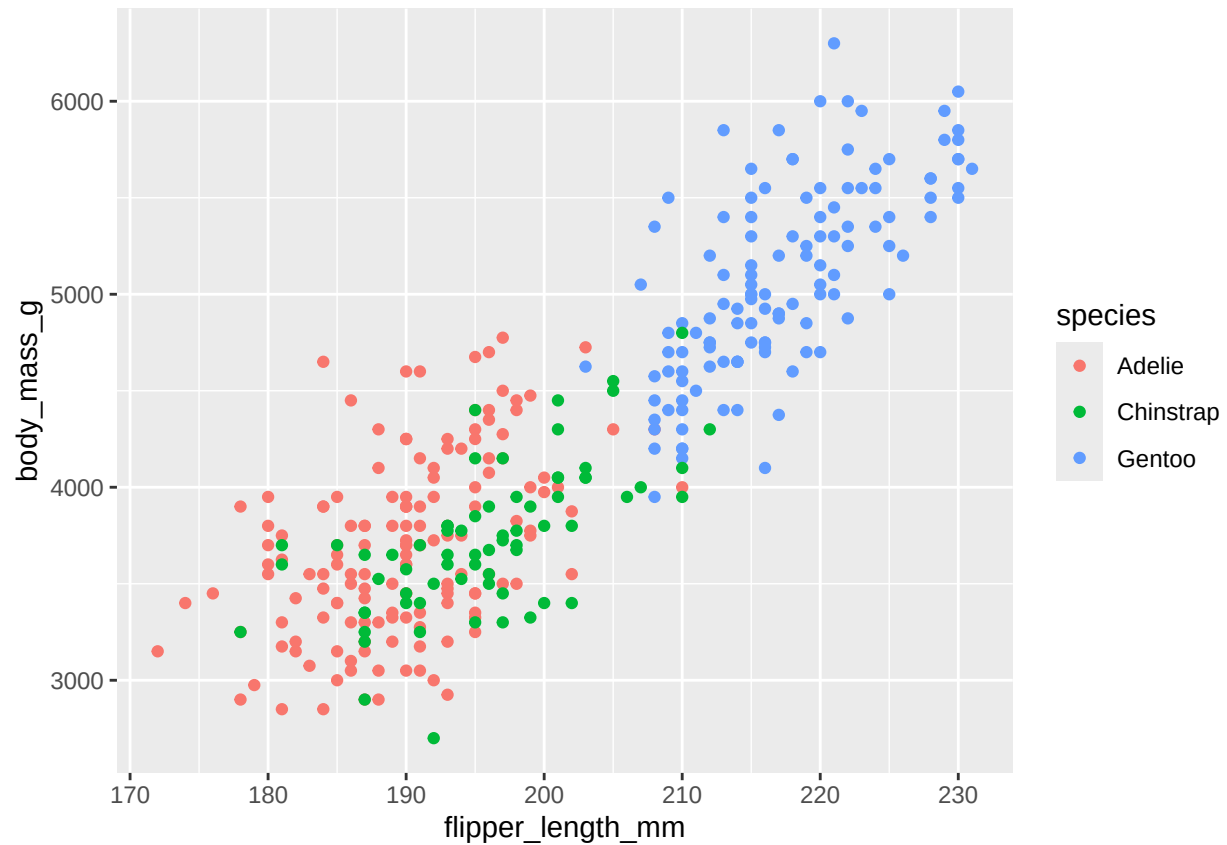
```
ggplot(  
  data = penguins,  
  aes(  
    x = flipper_length_mm,  
    y = body_mass_g,  
    color = species)) +  
  geom_point()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
ggplot(
  penguins,
  aes(
    x = flipper_length_mm,
    y = body_mass_g,
    color = species)) +
  geom_point()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Aesthetics across layers.

```
ggplot(
  data = penguins,
  aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
  geom_point() +
  geom_smooth(method = lm, se = FALSE)
```

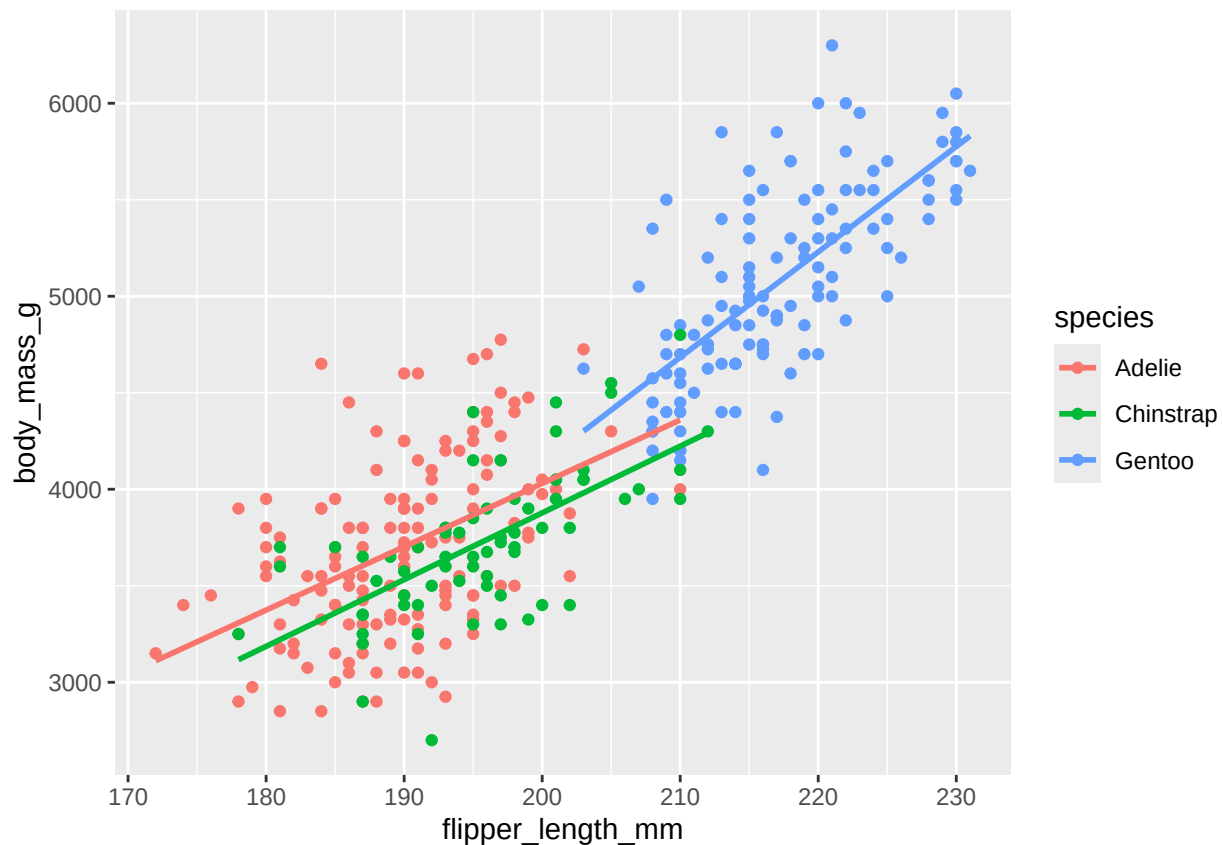
'geom_smooth()' using formula = 'y ~ x'

Warning: Removed 2 rows containing non-finite outside the scale range

('stat_smooth()').

Warning: Removed 2 rows containing missing values or values outside the scale range

('geom_point()').



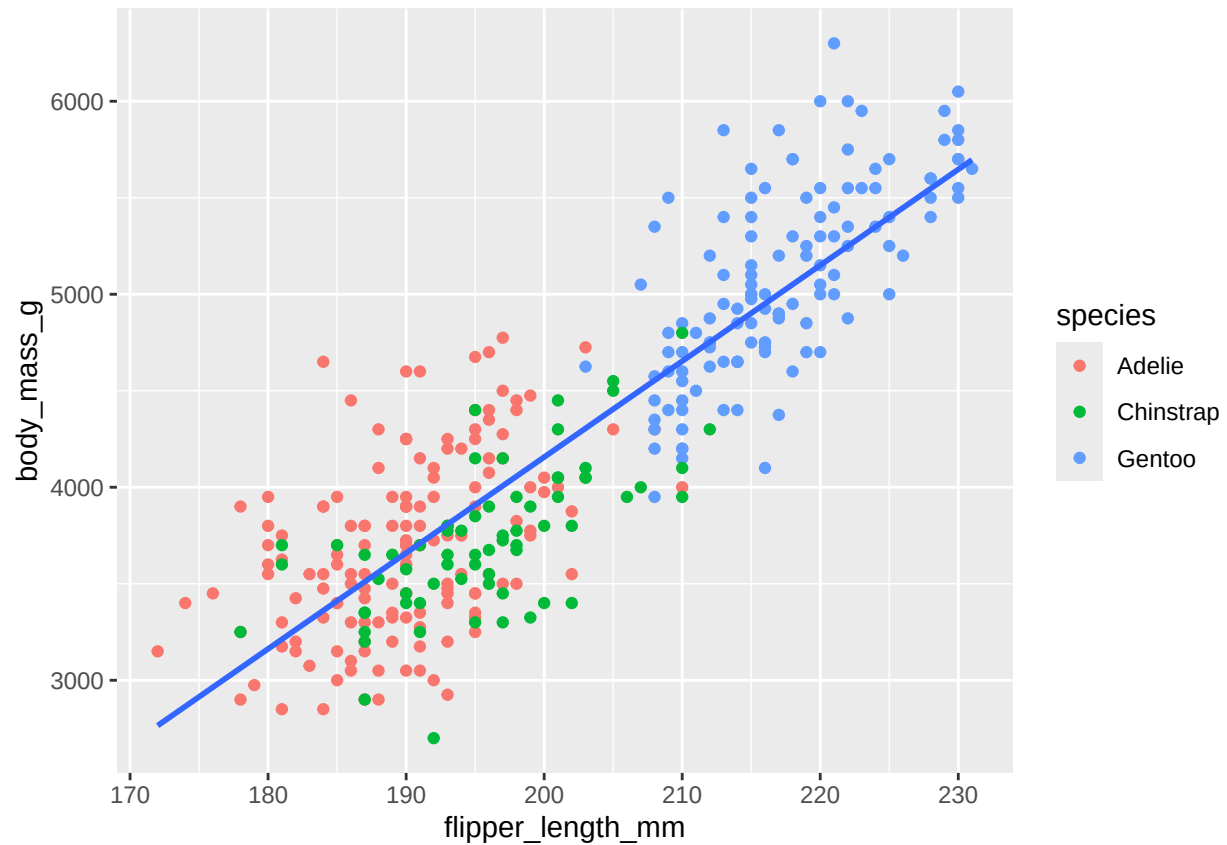
- 11) What does the option `se = FALSE` do? (Remove it and run and see what happens!)
- 12) What does the option `method = lm` do? (Remove it and run and see what happens!)
- 13) Be sure to undo both the changes above so that it reads `geom_smooth(method = lm, se = FALSE)` again. Now compare that plot above to this one:

```
ggplot(
  data = penguins,
  aes(x = flipper_length_mm, y = body_mass_g)) +
  geom_point(aes(color = species)) +
  geom_smooth(method = lm, se = FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```

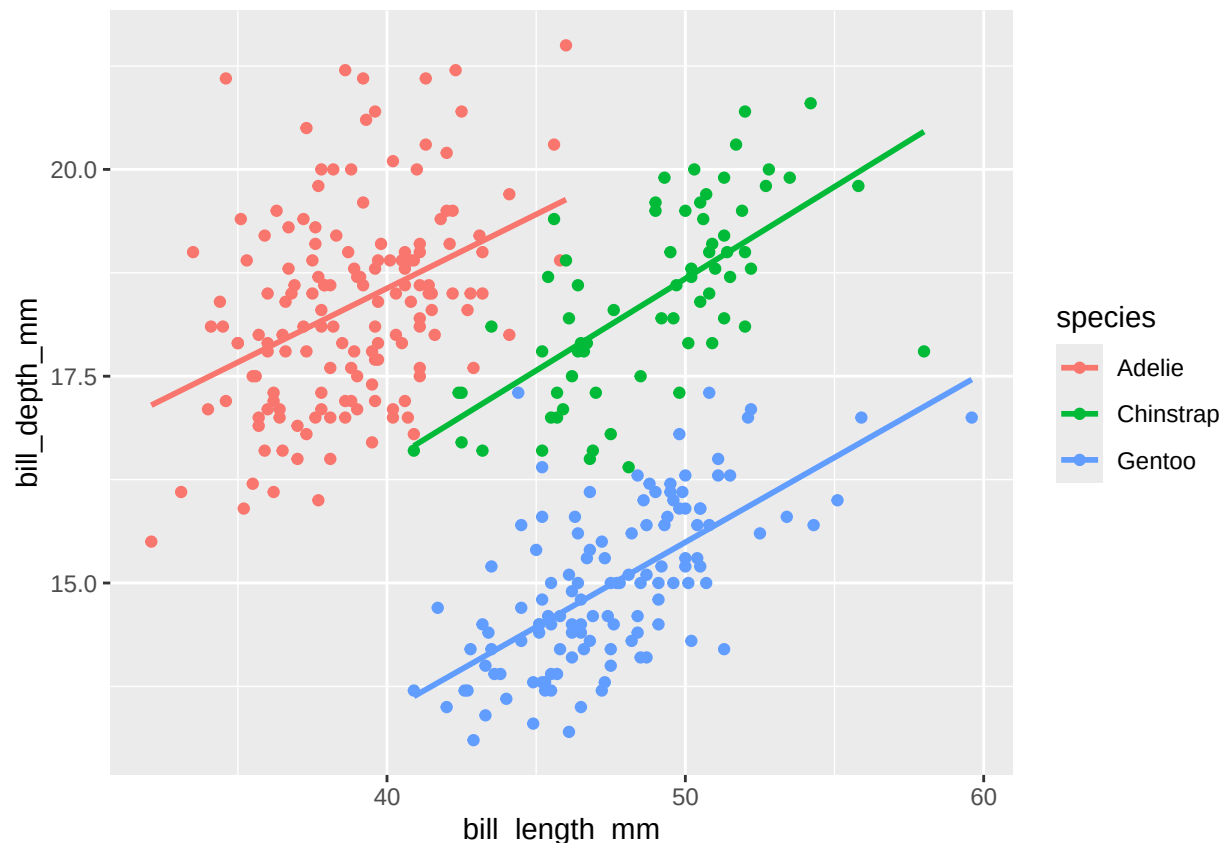


Describe the change in the graph's appearance. What change in the code led to this change in the plot?

More practice.

14) Write code to create the plot below.

To insert an R chunk here, you can use Ctr+Alt+I, the green chunk button, OR Code > Insert chunk.



Practice with figuring out errors!

Fixing errors in your code is a useful skill! If you get an error message, first read the entire message to see if it provides clues about what the error is or where it occurred.

Sometimes the messages are pretty cryptic, particularly when you are just learning R. Hopefully, it can give you a sense of at least WHERE the problem is.

If the message doesn't point you in a particular direction, just start at the beginning and carefully compare to example code that you know works properly.

- 15) For each of the graphs below, explain why we get errors (or they don't work as anticipated). Hint: Copy the code to a new chunk so that you can keep the original code for reference. Run one graph at a time with Ctrl+Enter (PC) or Command+Enter (Mac). Edit the code to make it work.

```
# Example 1
ggplot(data = penguins) +
  geom_point(
    aes(
      x = flipper_length_mm,
      y = body_mass_g,
      shape = 21,
      size = 5
    )
  )
)
```

```

# Example 2
ggplot(data = penguins)
+ geom_point(
  aes(x = flipper_length_mm, y = body_mass_g),
  shape = 21,
  size = 5
)

# Example 3
ggplot(data = penguins) +
  geom_point(
    x = flipper_length_mm,
    y = body_mass_g,
    shape = 21,
    size = 5
  )

# Example 4
ggplot(data = penguins) +
  geom_point(
    aes(
      x = flipper_length_mm, y = body_mass_g
    ),
    color = species,
    size = 5
  )

# Example 5
ggplot(data = penguins) +
  geom_point(mapping = aes(x = flipper_length, y = body_mass_g))

```